

Statistical Inference for Relative Risk

Winter 2026 DRP

Motivation

- ▶ We study how exposure (e.g., a drug) influences the risk of a later outcome.
- ▶ Both exposure and outcomes can be beneficial or harmful.

Notation

E is the event that an individual is exposed to a treatment

NE is the event that an individual is not exposed (note that $NE = E^c$)

D is the event that an individual gets a disease

Define $\pi_E = P(D \mid E)$ and $\pi_{NE} = P(D \mid NE)$.

What is a quantity that we can use to compare π_E against π_{NE} ?

“Relative risk” and “Odds ratios” are quantities that both compare π_E against π_{NE} , and are widely used in biostatistics, especially in epidemiology, public health, genetics, etc.

Relative Risk RR

$$RR = \frac{P(D | E)}{P(D | NE)} = \frac{\pi_E}{\pi_{NE}}$$

Interpretation

- $RR = 1 \rightarrow$ No association
- $RR > 1 \rightarrow$ Higher risk with exposure
- $RR < 1 \rightarrow$ Lower risk with exposure

Example:

- $RR = 3 \rightarrow$ probability of disease is 3 times higher if exposed

Empirical relative risk

If I have observed data:

		Disease	
		yes	no
Exposed	yes	a	b
	no	c	d

Then I can *estimate* RR risk with

$$\hat{RR} = \frac{\hat{\pi}_E}{\hat{\pi}_{NE}} = \frac{\frac{a}{a+b}}{\frac{c}{c+d}}$$

Odds Ratio OR

In addition to relative risk (RR), we can also compare π_E against π_{NE} with the odds ratio (OR):

$$OR = \frac{\frac{\pi_E}{1 - \pi_E}}{\frac{\pi_{NE}}{1 - \pi_{NE}}}$$

Sometimes there are technical reasons to prefer OR to RR , but outside the scope of today.

Interpretation

- Compares **odds** (not probabilities) of disease
- $OR = 1 \rightarrow$ No association

Note: In this presentation, we focus on Relative Risk

Statistical inference

Now, we will talk about *statistical inference*: using a sample to make a claims about a population.

There are two common approaches to conducting inference:

Hypothesis Test

- $H_0 : \text{RR} = 1$
- $H_A : \text{RR} \neq 1$

(Conducting a test often relies on the calculation of a p -value.)

Confidence Intervals

With a confidence level α , create an interval CI^α so that the true RR falls inside of CI^α , $100(1 - \alpha) \%$ of the time.

We will focus on this technique.

Confidence intervals for relative risk

Previously, I introduced an *estimate* of relative risk, which may be different from the true relative risk.

$$\text{Estimate: } \hat{RR} = \frac{\hat{\pi}_E}{\hat{\pi}_{NE}}$$

$$\text{True: } RR = \frac{\pi_E}{\pi_{NE}}$$

Research Question: How do we compute CIs for RR? Can we form a CI with $\hat{RR} \pm$ (margin of error)? Can we form a CI another way?

Delta method

To build a confidence interval, can we obtain a result such as:

$$\hat{R}R \approx N(RR, \text{Var}(\hat{R}R))$$

If so, we can build a 95% Confidence Interval as follows:

$$\hat{R}R \pm 1.96 * \sqrt{\text{Var}(\hat{R}R)}$$

We can establish this result using **Delta Method**

Simple version of the delta method

- * X_n is approximately distributed as $N(\mu_X, \text{Var}(X_n))$
- * Y_n is approximately distributed as $N(\mu_Y, \text{Var}(Y_n))$
- * $g(x, y)$ is a “smooth” function

The random variable $g(X_n, Y_n)$ is approximately distributed as $N(g(\mu_X, \mu_Y), \sigma_g^2)$, where

$$\sigma_g^2 = \nabla g^\top \Sigma \nabla g$$

$$\text{where } \nabla g = \begin{bmatrix} \frac{\partial g}{\partial x}(\mu_X, \mu_Y) \\ \frac{\partial g}{\partial y}(\mu_X, \mu_Y) \end{bmatrix}, \Sigma = \begin{bmatrix} \text{Var}(X_n) & \text{Cov}(X_n, Y_n) \\ \text{Cov}(Y_n, X_n) & \text{Var}(Y_n) \end{bmatrix}$$

Applying the delta method to relative risk:

$\hat{\pi}_E$ and $\hat{\pi}_{NE}$ are approximately normally distributed and independent, we have

$$RR = g(\pi_E, \pi_{NE}) = \frac{\pi_E}{\pi_{NE}}, \text{ and so } Var(\hat{RR}) \approx \nabla g^T \Sigma \nabla g$$

$$\text{Where } \Sigma = \begin{bmatrix} Var(\hat{\pi}_E) & 0 \\ 0 & Var(\pi_{\hat{NE}}) \end{bmatrix}, \nabla g = \begin{bmatrix} \frac{1}{\hat{\pi}_E} \\ \frac{\pi_E}{\pi_{\hat{NE}}^2} \end{bmatrix}$$

$$\text{We arrive at a formula: } Var(\hat{RR}) \approx \frac{Var(\hat{\pi}_E)}{\pi_{\hat{NE}}^2} + \frac{\hat{\pi}_E^2}{\pi_{\hat{NE}}^4} Var(\pi_{\hat{NE}})$$

Simulation Design

Is family history a risk factor for breast cancer?

		Diagnose breast cancer	
		yes	no
Family History	yes	432	87,787
	no	1,301	14,218

<https://pubmed.ncbi.nlm.nih.gov/28578505/>

$$\hat{\pi}_E = \frac{432}{432 + 87787}$$

$$\hat{\pi}_{NE} = \frac{1301}{1301 + 14218}$$

$$RR_{\text{simulation}} = \frac{\hat{\pi}_E}{\hat{\pi}_{NE}} = 17.12$$

Repeat the simulation 1000 times



Simulate dataset ($n = 10,000$)

- Exposure (family history)
- Outcome (breast cancer)



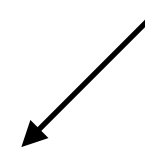
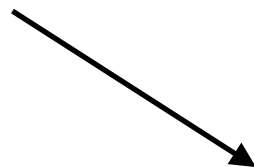
Estimate $\hat{RR}$

Delta Method CI

- Use gradient and compute variance
- Build a normal approximated interval

Bootstrap CI

- Bootstrap the simulated sample 2000 times
- Each time compute a bootstrap $\hat{RR}$
- Build an empirical confidence interval



Record Coverage & CI widths per simulation



Aggregate simulation results:

- Compute Empirical coverage and Average CI width

Simulation Results

Parameters:

- Number of Replicates: 1000 | Sample Size: 10000 people | $\alpha = 0.05$

Method	Coverage	Mean CI Width
Bootstrap Method (B = 2000)	95.0%	13.18
Delta Method	94.6%	12.34

- Delta Method: More precise, computationally cheap, but relies on normality assumption
- Bootstrap: Exact coverage, assumption-free, but wider intervals and more computational intensive

Thank you!